

Drosophila medulla neurons have increased order three motif three isomorphisms.

Words: 1329

1. Methods

1.1. *Drosophila* Data

The *Drosophila* connectome data was collected and published by Scheffer et al. [1] This data was downloaded and investigated using the region of interest file.

The data were explored using the CSV ROI Filtering module. Running this file produces a list of how many edges exist for a given region of interest. The file as is selects for data in the medulla (labelled "ME(R)") and exports this selection to a CSV file for ease of computation in other modules. ME(R) data was selected to ensure the programme could run in a timely manner on my hardware.

1.2. Identifying Motifs

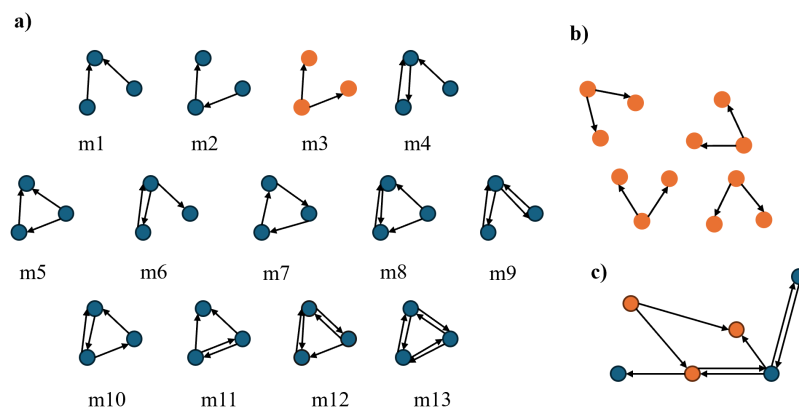


Figure 1. A) An illustration of all possible motifs which have three nodes. The motif number is beneath the motif it represents. Motif three is depicted in orange. B) Motifs can appear in any orientation. Graphs which have the same motif are considered isomorphic. All graphs in B are isomorphic. C) An example network, illustrating motif three. Some nodes share have other connections, but the relationship between the three orange nodes is isomorphic to motif 3.

The motifs were analysed using the ME(R) Analysis module, which requires that the ME(R) data exists as a CSV file and that the motifAnalysis module in the code environment. Running this programme draws a directed graph of connections in the medulla, identifies each possible isomorphism (see Figure 1), and plots the number of each isomorphism on a histogram called a motif spectrum.

1.3. Null Graph

A null graph is a model graph generated to compare experimental data against a control [2]. Usually these graphs are random and display matched characteristics. In this case, the number of nodes and the density of the graph will remain the same.

Random graphs were generated using an Erdős-Rényi model in the normalisingMotifs module, which runs independently from the main programme. One hundred random graphs were generated and analysed for the number of occurrences of the motif of interest, which was exported to a CSV file for use in the main programme due to the time required to run this module.

The occurrences of the motif of interest in each graph were compiled into a histogram in the main programme to create a normalised distribution of the number of its isomorphic graphs, based on the law of large numbers. Identifying where the ME(R) data falls with relation to the distribution indicated how unusual the value versus random graphs with similar characteristics.

2. Results

2.1. Data Visualisation

Immediately after processing the data, the entirety of the data was visualised using NetworkX in Figure 2. Intriguingly, the graph has multiple components. A quick visual inspection indicates that nodes 35 and 36 form their own component as well as 67 and 68. This suggests that the *Drosophila* medulla is involved in circuits which aren't connected to the rest of the network; therefore these isolated components likely are involved in different functions or are connected through neurons not represented in the medulla. While this isn't fully explored in this paper, it could be an avenue for future investigation.

Descriptive statistics about the ME(R) data can be found in Table 1 which are used to make a null graph as described in 1.3. Of particular note is the density. Because of the function of the medulla as a thin band of fibres which connects the spinal cord and the rest of the brain, it makes sense that the ME(R) is more sparse than dense. Further research could be done to compare the density of the medulla versus other regions to validate this assertion.

Statistic	Value
Nodes	76
Edges	117
Density	0.0205

Table 1. Descriptive statistics about the *Drosophila* ME(R) Network. It lists the number of nodes and edges, as well as the density of edges within the graph.

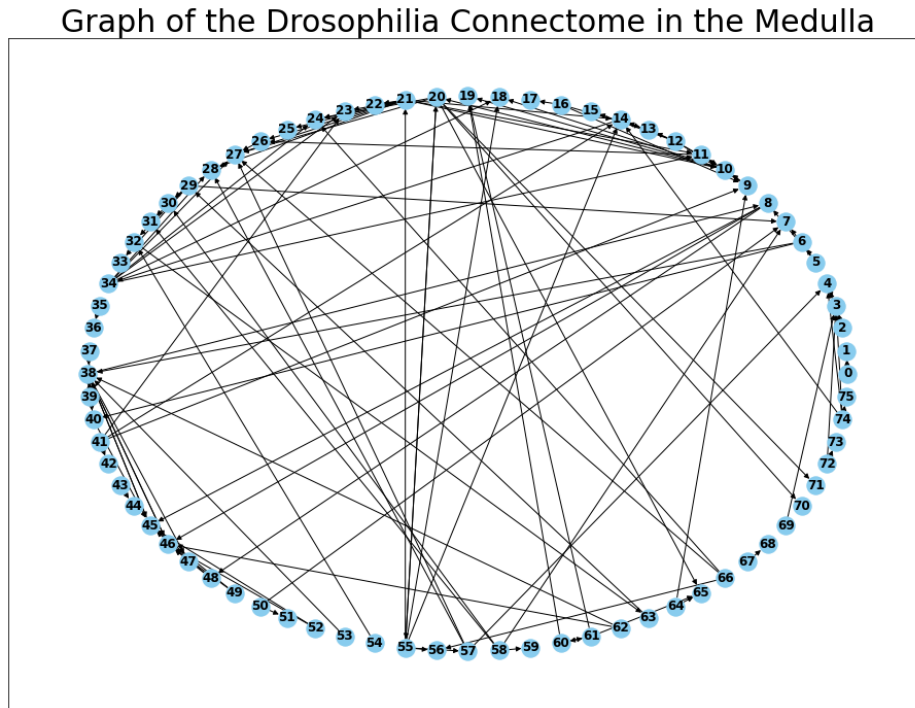


Figure 2. A directed graph of the *Drosophila* connectome data filtered for nodes in the medulla. Each node has been assigned a number based on its index in the edge list.

2.2. The Motif Frequency Spectrum

Analysing the data for the motif frequency revealed that of the 70300 possible triplet pairs in the graph, 357 of them were isomorphic to a three node motif meaning that 0.5% of the triplets were isomorphic to a node 3 motif. This suggests that a majority of these triplets are not connected to each other or only have two nodes connected to each other.

The motif frequency spectrum was visualised on a histogram (see Fig. 3). Isomorphisms of motif 3, 1, and 2 appeared most frequently, in that order. Motif 3 accounts for 39% of all the isomorphic graphs. The frequency of the first three motifs indicates that this is a sparse graph, with fewer reciprocal connections such as those seen in motifs 9-13 (see Fig. 1). This is likely related to the sparseness of the graph, as the first three motifs have fewer paths connecting them.

To ensure, identify a motif of interest, a singular random graph was generated to produce a motif frequency spectrum (see Appendix 1). These two spectra were imposed on each other to see which motifs were different in the random graph and the ME(R) graph (see

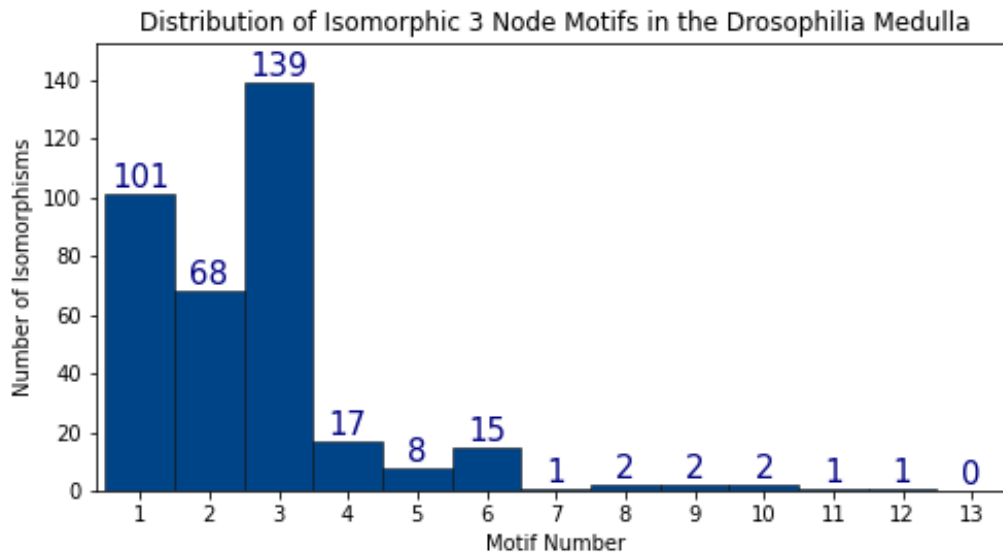


Figure 3. The motif spectrum of the medulla connectome data. The total times an isomorphism was identified is on top of each bar and the motif number is on the x axis.

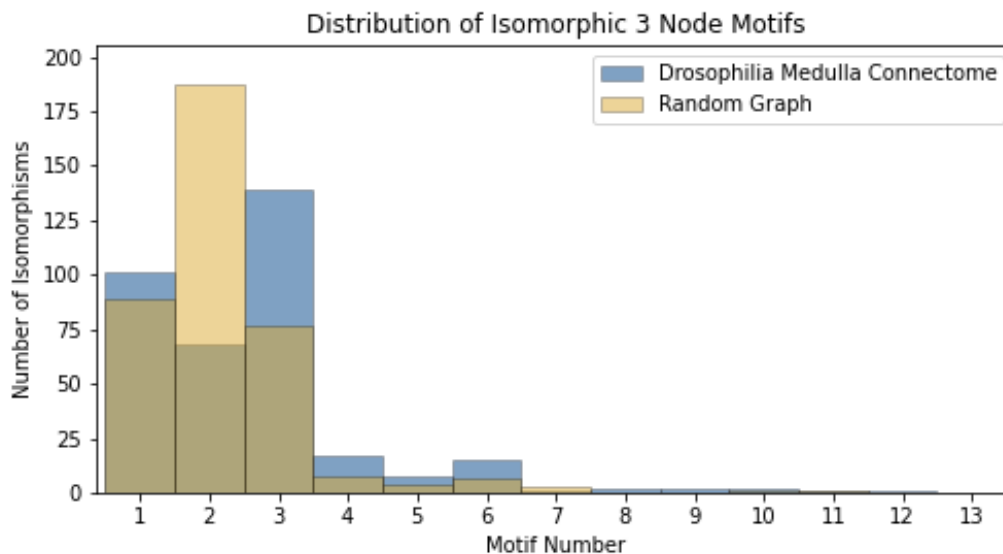


Figure 4. The motif spectra of the ME(R) connectome data and the randomly generated data were superimposed on each other to visually identify any potential differences between the data sets..

Fig. 4). Visual inspection suggested that motifs 1-6 could be worth investigating.

Motif three was selected as the region of interest, since isomorphisms of motif 3 appeared most frequently in the *Drosophila*. If the occurrences of motif three isomorphisms in the medulla are significantly greater than expected, it also could support the role of the medulla as a relay into higher brain regions as one node connects to two other nodes in a motif 3 configuration.

2.3. The Null Model

A null model was generated as described in Section 1.3 with motif 3 as the motif of interest. Of the 100 generated graphs, none of them had more than 130 occurrences of motif 3, as seen in the ME(R) data (see Fig. 5. Most graphs had between 60-90 occurrences of motif 3 subgraphs. Based on Fig. 5, one can conclude that motif 3 appears significantly more in the medulla data than similar graphs, which implies that motif three is over represented in the medulla.

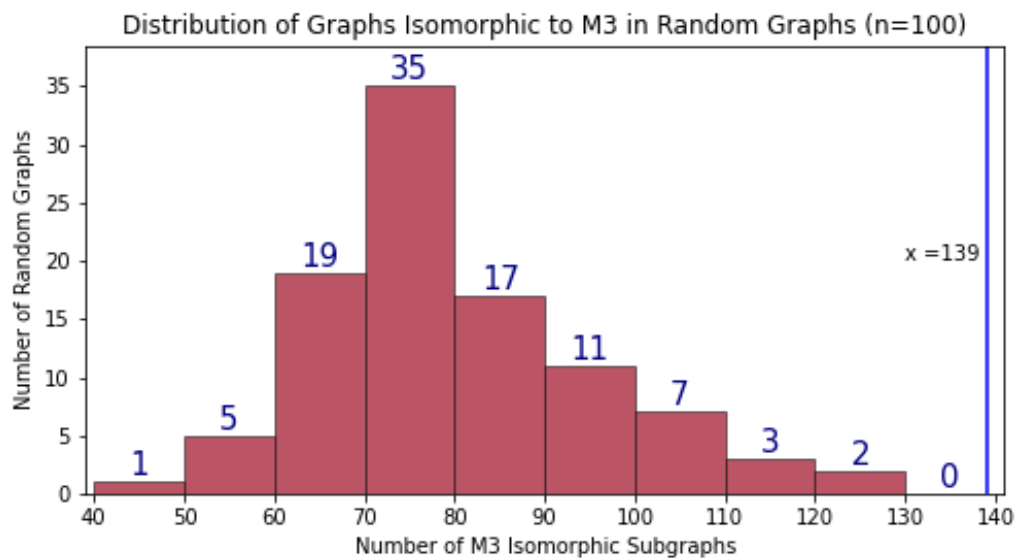


Figure 5. A histogram indicating distribution of the third motif in 100 random Erdős-Rényi graphs with $n=76$ and density = 0.0205. The line $x = 139$ shows where the ME(R) data falls with respect to the distribution.

3. Appendix

The random graph used for the initial comparison of motifs were also visualised as both a graph (see Fig. 6 and a motif frequency spectrum (see Fig. 7). The randomised graph seems to have more connections than the medulla data, though this could be because more nodes have connections to nodes across the circle. It could appear more similar to the ME(R) data in a different arrangement. The other note of interest is that this graph appears to be entirely connected and lacks individual components. This differs from the ME(R) data, which had a few nodes which only connected to themselves in the medulla. The motif frequency spectrum seems to share many surface level traits with the ME(R)

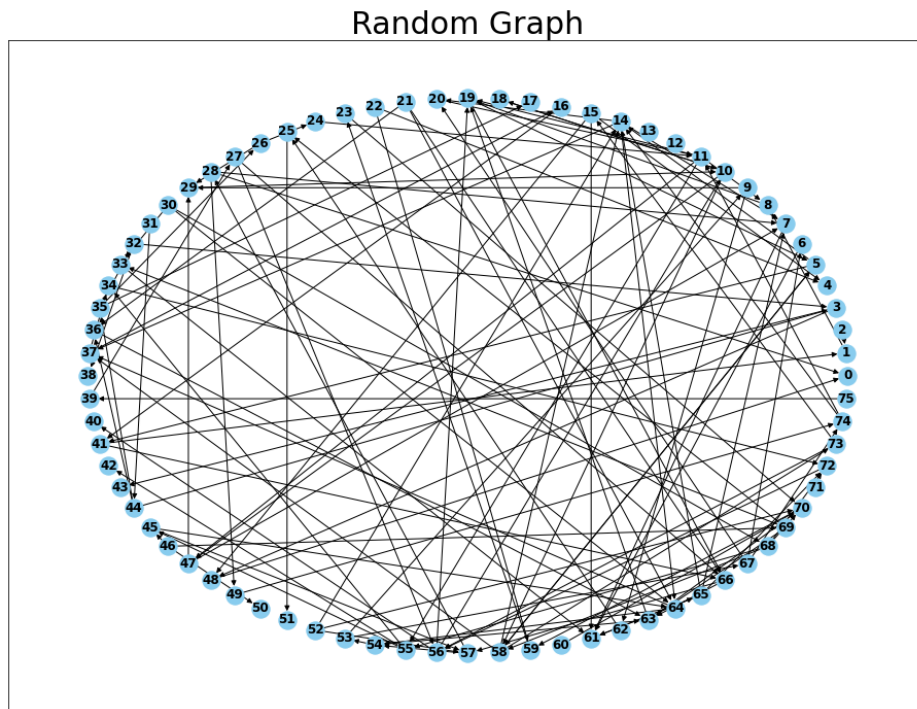


Figure 6. A random graph using the node count and density of the ME(R) data.

data, namely that most isomorphic motif subgraphs were concentrated in motifs 1-6. Motifs 7-13 seems to be underrepresented in both types of graphs. These insights may not hold across all randomised graphs, as this is only a singular randomised graph. Further investigation would be required to see if these traits hold across all similarly generated graphs, which is outside the scope of this paper.

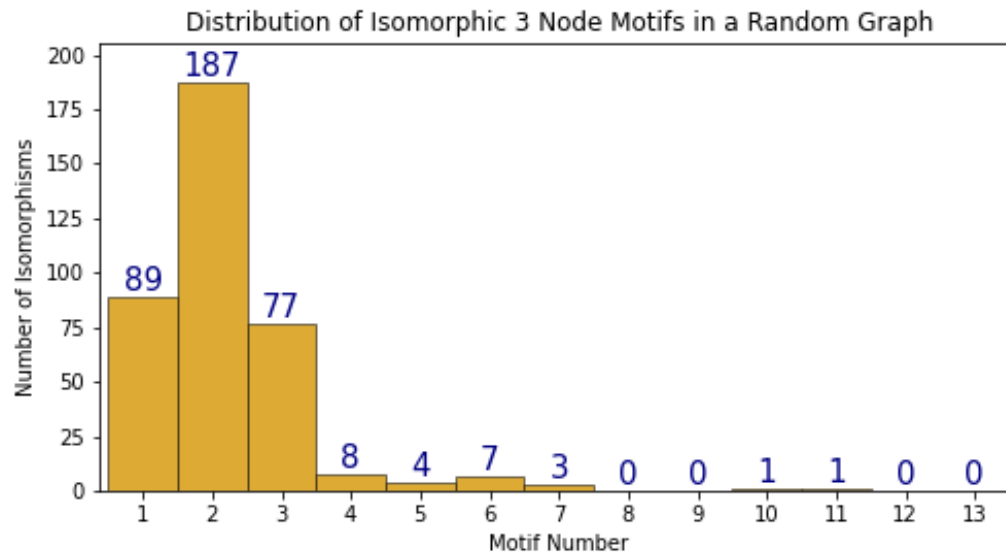


Figure 7. The motif spectrum of a randomised graph using the density and node count of the ME(R) data. The total times an isomorphism was identified is on top of each bar and the motif number is on the x axis.

References

- [1] Scheffer L K, Xu C S, Januszewski M, Lu Z, Takemura S y, Hayworth K J, Huang G B, Shinomiya K, Maitlin-Shepard J, Berg S, Clements J, Hubbard P M, Katz W T, Umayam L, Zhao T, Ackerman D, Blakely T, Bogovic J, Dolafi T, Kainmueller D, Kawase T, Khairy K A, Leavitt L, Li P H, Lindsey L, Neubarth N, Olbris D J, Otsuna H, Trautman E T, Ito M, Bates A S, Goldammer J, Wolff T, Svirskas R, Schlegel P, Neace E, Knecht C J, Alvarado C X, Bailey D A, Ballinger S, Borycz J A, Canino B S, Cheatham N, Cook M, Dreher M, Duclos O, Eubanks B, Fairbanks K, Finley S, Forknall N, Francis A, Hopkins G P, Joyce E M, Kim S, Kirk N A, Kovalyak J, Lauchie S A, Lohff A, Maldonado C, Manley E A, McLin S, Mooney C, Ndama M, Ogundeyi O, Okeoma N, Ordish C, Padilla N, Patrick C M, Paterson T, Phillips E E, Phillips E M, Rampally N, Ribeiro C, Robertson M K, Rymer J T, Ryan S M, Sammons M, Scott A K, Scott A L, Shinomiya A, Smith C, Smith K, Smith N L, Sobeski M A, Suleiman A, Swift J, Takemura S, Talebi I, Tarnogorska D, Tenshaw E, Tokhi T, Walsh J J, Yang T, Horne J A, Li F, Parekh R, Rivlin P K, Jayaraman V, Costa M, Jefferis G S, Ito K, Saalfeld S, George R, Meinertzhagen I A, Rubin G M, Hess H F, Jain V and Plaza S M 2020 *eLife* **9** e57443 ISSN 2050-084X publisher: eLife Sciences Publications, Ltd URL <https://doi.org/10.7554/eLife.57443>
- [2] 2016 Chapter 10 - Null Models *Fundamentals of Brain Network Analysis* ed Fornito A, Zalesky A and Bullmore E T (San Diego: Academic Press) pp 355–381 ISBN 978-0-12-407908-3 URL <https://www.sciencedirect.com/science/article/pii/B9780124079083000108>